26900042 **Reg No:……....**

Name :……..

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

FIFTH SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCBIF300 - PERL & BIOPERL PROGRAMMING

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

PART A

Short Answer Questions

Answer any 5 questions

Each question carries 2 Marks

1. Define an array element in Perl. *[K] / [CO1]*

- An individual scalar value stored within an array structure, accessed using a specific numerical index starting at 0.
- It is referenced using the scalar sigil $ instead of @ (e.g., $array[0]).

2. Mention the operator used to read data from a file. *[K] / [CO1]*

- The line input operator, also known as the diamond operator <> or <FILEHANDLE>.
- It reads a single line from an open file handle in a scalar context or all lines in a list context.

3. State the year in which Perl was originally developed. *[K] / [CO1]*

- Perl was originally created and released by Larry Wall in 1987.

4. What is meant by finding all occurrences of a pattern in Perl? *[K] / [CO3]*

- Searching a string globally to match every single instance of a given regular expression pattern.
- Achieved using the global modifier g at the end of a match expression (e.g., m/pattern/g).

5. Show the general syntax of a Perl built-in function. *[K] / [CO4]*

- function_name(argument1, argument2, ...);
- Parentheses can often be omitted if the function has been predefined or built-in (e.g., print "Hello";).

6. Apply BioPerl to retrieve a GenBank record and obtain its sequence identifier. *[A] / [CO6]*

```
use Bio::DB::GenBank;
$db = Bio::DB::GenBank->new();
```

```
$seq_obj = $db->get_Seq_by_id('X77802');
print $seq_obj->display_id(); # Obtains sequence identifier
```

7. Apply the appropriate BioPerl module to create a simple BioPerl program. *[A] / [CO6]*

```
use Bio::Seq;
# Creates a simple sequence object
$seq = Bio::Seq->new(-seq => 'ATGGCCATT', -alphabet => 'dna', -id => 'seq1');
print "ID: ", $seq->id(), " Length: ", $seq->length();
```

[2x5 = 10]

PART B

Short Essay Questions

Answer any 4 questions

Each question carries 5 Marks

8. Explain the purpose of the .pl file extension in Perl programming. *[K] / [CO1]*

- Serves as the standard file extension to designate a plain text file containing executable Perl code.
- Helps the operating system and development environments recognize and properly associate the script with the Perl interpreter.
- Differentiates execution scripts from reusable script libraries or modules which typically use the .pm extension.
- Enhances code organization and portability across different command-line environments.

9. Describe the working of the for loop and foreach loop in Perl. *[U] / [CO2]*

- for loop: Typically used for traditional, index-based numeric counters with explicit initialization, condition, and re-evaluation expressions.
- foreach loop: Designed to iterate seamlessly through every individual element of an array or a list from beginning to end.
- Aliasing: In a foreach loop, the loop control variable acts as a direct alias to the current element; modifying it alters the original array element.
- Interchangability: In Perl, the keywords for and foreach are technically interchangeable, but stylistic conventions prefer choosing them based on readability.

10. Explain the purpose of the while loop in Perl. *[U] / [CO2]*

- Repeatedly executes a block of code as long as a specified test condition evaluates to true.
- Evaluates the condition expression at the very top of the loop structure before entering or repeating the block.
- Commonly utilized for reading input data streams line-by-line until the End-Of-File (EOF) marker is reached.
- Requires the state of the loop condition to change inside the execution block to prevent creating an infinite loop scenario.

11. Describe how wildcard-like patterns can be represented using regular expressions in Perl. *[K] / [CO3]*

- Period metacharacter (.): Matches any single character except a newline sequence.
- Asterisk quantifier (*): Matches zero or more repetitions of the immediately preceding character or subpattern.
- Plus quantifier (+): Matches one or more repetitions of the immediately preceding character or subpattern.
- Dynamic combinations: Combining these elements (such as .* or .+) behaves identically to traditional command-line global wildcards.

12. Generate a suitable method for designing subroutines that accept arguments and return values in Perl programming. *[C] / [CO5]*

- Define the subroutine using the sub keyword followed by a descriptive block identifier.
- Retrieve passed incoming arguments within the block by unpacking the special default array variable @_.
- Use the explicit return keyword to pass computed values back out to the calling function or main execution scope.

```
sub add_numbers {
    my ($num1, $num2) = @_;
    my $sum = $num1 + $num2;
    return $sum;
}
```

13. Use Bio::AlignIO to read an alignment and obtain the sequence information from the alignment. *[A] / [CO6]*

- Initialize a Bio::AlignIO streaming input object by defining the target filename and format parameter.
- Use the next_aln() method to pull alignment records systematically from the initialized file stream.
- Iterate over individual sequence objects contained within the alignment layer via the each_seq() method to read information.

```
use Bio::AlignIO;
$in = Bio::AlignIO->new(-file => "align.fasta", -format => "fasta");
while (my $aln = $in->next_aln()) {
    foreach my $seq ($aln->each_seq()) {
        print "ID: ", $seq->id(), " Sequence: ", $seq->seq(), "\n";
    }
}
```

[5x4 = 20]

PART C

Long Essay Questions

Answer any 2 questions

Each question carries 10 Marks

14. Explain operator precedence and associativity in Perl with suitable examples. *[K] / [CO1]*

- Operator Precedence: Determines the parsing priority and execution order when distinct operators are placed together in a single statement. (3 Marks)
- Associativity: Specifies the direction of evaluation (left-to-right or right-to-left) when multiple operators sharing identical precedence appear together. (3 Marks)
- Precedence Example: In 5 + 2 * 3, multiplication (*) holds a higher precedence ranking than addition (+), computing to 11 instead of 21.
- Associativity Example: The assignment operator (=) evaluates right-to-left, allowing chain operations like $a = $b = 10.

15. Explain the procedure for executing a Perl script from the command line with suitable examples. *[U] / [CO1]*

- Direct Execution Method: Invoke the script file directly by passing it as an explicit argument to the command-line perl interpreter command. (3 Marks)
- Executable Script Method: Grant executable permissions to the script file and define the specific path to the Perl binary engine on the first line. (3 Marks)
- Interpreter call: perl script.pl
- Executable setup (Unix/Linux): Run chmod +x script.pl and ensure the script starts with the shebang declaration line: #!/usr/bin/env perl. Execute using ./script.pl.

16. Explain how character sets can be used for pattern matching in Perl. Illustrate your answer with different examples. *[K] / [CO3]*

- Character Classes Definition: Enclosing characters inside square brackets [] to match any single character belonging to that specified group. (3 Marks)
- Ranges and Negation: Using a hyphen - to define a consecutive character block, or an anchor caret ^ at the start to negate the entire match criteria. (3 Marks)
- [A-Z]: Matches any uppercase alphabetic character.
- [^0-9]: Negates digits, matching any non-numeric character.
- [atgcATGC]: Matches valid DNA nucleotides regardless of case sensitivity constraints.

17. Identify the different types of anchors used in Perl pattern matching and state the purpose of each. *[K] / [CO3]*

- Definition of Anchors: Special regular expression metacharacters that do not match individual text characters, but instead lock matches to explicit positions within strings. (2 Marks)
- Caret Anchor (^): Forces a successful pattern match exclusively at the absolute starting boundary position of the evaluation string. (2 Marks)

- Dollar Anchor ($): Compels a pattern match to align precisely with the absolute final boundary position or right before a trailing newline of a string. (2 Marks)
- Word Boundaries (\b and \B): \b asserts a boundary position between an alphanumeric word character and a non-word character space; \B negates this check. (2 Marks)
- Example 1: if ($str =~ /^ATG/) checks if a sequence string explicitly initiates with a start codon.
- Example 2: if ($str =~ /TAA$/) checks if a sequence string explicitly terminates with a stop codon.

[10x2 = 20]